

Nicolas Moreira Ferreira

+55 13 99794-9634 | São Vicente, São Paulo, Brazil | nikola.snay@hotmail.com | github.com/nicolasmoreiraferreira | linkedin.com/in/nicolasmoreiraferreira | nicolasmoreiraferreira.github.io/portfolio

SUMMARY

Full stack developer building and running web applications end to end: React and TypeScript interfaces, typed APIs, relational databases and automated tests. I maintain a financial dashboard in production and a desktop automation platform with 190+ Python modules, treating accessibility, testing and CI/CD as requirements.

TECHNICAL SKILLS

- **Front end:** HTML5, CSS3, JavaScript (ES2022+), TypeScript, React, Vite, Tailwind CSS, responsive design, accessibility (WCAG)
- **Back end & data:** Node.js, Express, tRPC, REST APIs, MySQL, Drizzle ORM, Zod, Python
- **Testing & quality:** Vitest, Testing Library, Playwright, ESLint, TypeScript strict mode
- **Tools & DevOps:** Git, GitHub, GitHub Actions (CI/CD), PyInstaller, browser extensions (Manifest V3)

PROJECTS

Full Stack Developer, Financial Dashboard (controlefinanceirosnay.com) Jul 2026 — Present

- Built a typed end-to-end API with tRPC over Express, persisting to MySQL through Drizzle ORM with versioned migrations and Zod validation.
- Implemented expense entry by WhatsApp message: a webhook parses amount, category and card from free text, and writes are idempotent by message identifier, so a resend never duplicates a transaction.
- Kept domain logic in pure functions tested in isolation (invoice cycle, balance, budgets, goals), reaching 130+ automated tests with type checking and build on every delivery.
- Isolated data per account — nominal invitations, empty first dashboard, dedicated sync credentials. Code is private (real financial data); showcase: <https://github.com/nicolasmoreiraferreira/controle-financeiro>

Developer, BOTSNAVY — Web Process Automation Platform 2026 — Present

- Built a desktop platform running workflows on websites autonomously, with 190+ Python modules.
- Orchestrated isolated Playwright browser sessions for parallel execution without interference, handling threads, state persistence and recovery from partial failures.
- Packaged for Windows with a versioned release pipeline validated in GitHub Actions, plus licensing and automatic updates for non-technical end users.

Author, AI Agent Guardrails (github.com/nicolasmoreiraferreira/ia-guardrails) 2026 — Present

- Built the control layer that lets an AI agent write code in a production system without access to credentials, customer data or browser sessions: write blocked on sensitive paths before it happens, secret scanning on every approved write and independent reviewers required to return an explicit verdict.

Front-end Developer, Estados — UI States Lab (github.com/nicolasmoreiraferreira/estados) 2026 — Present

- Built an open-source React 19 and TypeScript app forcing twelve real API conditions — server error, expired session, no permission, no connection, corrupted payload, 5,000 records — so every screen can be verified on demand.
- Designed a single state renderer leaving no way to render content without handling loading, error and empty states, enforced by tests across all four screens.
- Covered it with 175 automated tests (103 unit, 72 browser) on desktop and mobile, with list virtualization keeping 20 rows in the DOM out of 5,004.

Full Stack Developer, Digital Invitation with RSVP Jul 2026 — Jul 2026

- Built an event site used by dozens of families, with real-time countdown and online RSVP separating adults from children, validated and stored server-side, plus an owner dashboard with consolidated totals.

EDUCATION & CERTIFICATIONS

Cruzeiro do Sul Virtual 2026 — 2028

Bachelor of Technology in Systems Analysis and Development (in progress) Distance learning, Brazil

Curso em Vídeo 2023

Front-End Development: HTML5, CSS3 and JavaScript Online, Brazil

Digital Universe and AI Explorer, Cruzeiro do Sul University 4h · Oct 2026

Hardware and software, algorithms, internet and cloud and AI ethics · Verification code: a2fb632a-1477-480f-9cec-05a4f753d802

LANGUAGES

Portuguese: Native · **English:** Basic — I read technical documentation and follow written communication; conversation is not yet fluent.